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On a fundamental level, the majority of biological processes are governed by the ability of proteins to recognize each other and small molecule ligands in a specific manner. One might take as an example the signal transduction pathway involved in bone resorption. It is known that many of the cytokines involved in bone resorption (such as IL-6 and IL-11) are recognized by specific cell surface receptors and that those receptors are responsible for initiating a cascade of protein-protein interactions (the JAK/STAT pathway) which ultimately increases the activity of the STAT transcriptional activator. In considering any single protein, it is often the case that upon the binding of one ligand or protein a conformational change occurs which alters the binding profile for all other ligands and proteins. This is the basis of allosteric control. It is the goal of our project to screen proteins of interest against combinatorial libraries of peptides and other ligands so that we can identify those ligands which are capable of stabilizing conformational changes. We believe such identifications can be made through a series of ELISA experiments in which our target protein is screened in the presence and absence of small molecules which are known to effect conformational changes. The isolated library ligands

Task Description:	might then be applied as artificial allosteric regulators. In a proof-of-principle series of experiments we intend to study the enzyme 3-deoxy-D-manno-octulosonic acid 8-phosphate synthase (KDO 8-P synthase). KDO 8-P synthase from the extremophile Aquifex aeolicus was chosen because it does not undergo enzymatic turnover at room temperature, passes through a known conformational change upon binding to substrates or inhibitors within its active site, can be easily expressed in large quantities, and has a large library of inhibitors. Following our initial studies on KDO 8-P synthase, we will apply our methodology to systems with more direct biomedical relevance. Specifically, as alluded to earlier, one potential application for this work is in preventing pathological bone resorption, an important phenomenon associated with long-term spaceflight. Astronauts who have been in microgravity for several months are placed at an increased risk of suffering pathological fractures due to abnormal bone resorption. It is possible to envision a novel method for preventing bone resorption which would involve the development of allosteric ligands which bind with high affinity to STAT and prevent it from interacting with the other components of the JAK/STAT pathway.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Regulated transcription of specific genes is one of the fundamental processes that underlie all of human physiology. As a result, the mis-regulation of transcription is related to much of human pathophysiology as either a cause or a consequence. This realization has spurred intense interest in uncovering the fundamental characteristics of transcriptional regulation and in the eventual development of molecules capable of regulating transcription in living systems and acting as transcription-based therapeutics. It has been established that in any given eukaryotic cell there are more than 100 different proteins from a number of different functional classes involved in the proper regulation of transcription. We have developed a particular interest in studying two of the most fundamentally important types – transcriptional activators and transcriptional coactivators – and the interactions between them. The rational design of molecules for artificial transcriptional regulation will be greatly enhanced by a deeper understanding of activator-coactivator interactions.
Task Progress:	A thorough understanding of the surfaces targeted by transcriptional activation domains is necessary for the rational design of future generations of artificial transcriptional activators. Some of the best characterized activation domain binding surfaces are targeted by a number of different activation domains, implying that such surfaces have permissive binding profiles. This also suggests that a variety of peptidomimetic and organic small molecule scaffolds with appropriate functionality could target such surfaces and function as transcriptional modulators. In the work described below, we directly examined the permissiveness of one of the activation domain binding sites of the CBP and Med15 KIX domains by synthesizing two series of peptidomimetic analogs of two peptides known to target the CBP KIX domain site. Our results indicate that appropriate functionality can be arrayed in a variety of orientations while maintaining binding affinity for the target sites we examined. In 2000, Montminy and co-workers isolated two peptides from a phage display panning of a CBP KIX domain construct (mouse CBP(586-672)).12 The peptides that were thus obtained (KBP 1.66: SSLVDLILFGS and KBP 2.20: SWAVYELLFGS) possessed a degree of sequence similarity to endogenous activation domains (Figure 4), and upon fusion to the Gal4 DNA-binding and dimerization domains (Gal4(1-148)) KBP 1.66 and KBP 2.20 were found to function as artificial activation domains in HEK-293 cells. KBP 2.20 was observed to be the more potent of the two activation domains, commensurate with its higher in vitro binding affinity for the CBP KIX domain. Mutational analysis of the KIX domain indicated that KBP 1.66 and KBP 2.20 were binding a largely overlapping portion of the same hydrophobic groove on helices a1 and a3 that is targeted by such activation domains as those of CREB and c-Myb. Given the broad, shallow hydrophobic groove to which KBP 1.66 and KBP 2.20 bind, it is reasonable to assume that the functionality from the peptides might be ostensibly transferred to various peptidomimetic scaffolds that, though displaying the functional groups differently, might still retain binding affinity for the CBP KIX domain. Investigating the abilities of such ligands to interact with the CBP KIX domain will allow for a direct determination of the permissiveness of such activation domain binding surfaces. In providing fundamental insight into the requirements necessary for interacting with activation domain binding surfaces, such information would also provide new guiding principles for the design of future generations of artificial activation domains. To investigate the permissiveness of activation domain binding sites, a series of peptidomimetic scaffolds was chosen that, while maintaining the intrinsic ability to adopt helical conformations, still varied considerably in the manner in which they were able to display a given set of functionality. For example, d-peptides were chosen because they are known to fold into a-helices that are enantiomers of those formed by the analogous l-peptides and hence both types of ligands can array the same functionality along a single helix face. d-Peptides are also easily synthetically accessed with any of the amino acid side chains found in natural peptides. In any a-helix formed by either an l-peptide or a d-peptide, there will be 3.6 residues per complete turn of the helix leading to any three residue side chains being cofacial that have a 1, 4, 8 relationship to each other. Following the synthesis and fluorescence labeling of the KBP 1.66 and KBP 2.20 derived ligands, they were tested for binding to the CBP KIX domain using fluorescence polarization. Perhaps not surprisingly, the l- and d-peptide versions of KBP 1.66 were found to bind with very similar affinity to the CBP KIX domain, with KDs of 75 μM and 84 μM, respectively. The ß-peptide ligand demonstrated three-to-four fold better binding affinity (KD 20 μM) than the parent peptide, perhaps indicating a more ideal arrangement of the amino acid side chains necessary for binding to the KIX domain. Indeed, it is not possible to align 3 large, hydrophobic residues along a single face of an a-helical conformation of either the l- or d-peptide ligands . However, the adoption of a 14-helix by the ß-peptide would allow two isobutyl side chains from ß-homoleucine residues and one benzyl side chain from a ß-homophenylalanine residue to all reside on the same helical face. Interestingly, the KBP 1.66 derived peptoid did not display any detectable binding affinity for the KIX domain. Though the appropriate side chain functionality is obviously present in the peptoid, it may be that one or more backbone amide N-H bonds in KBP 1.66 made important contact with a residue or residues in the CBP KIX domain, and that loss of any such interactions is enough to abrogate binding. Further, the presence of only one aromatic side chain in the peptoid would severely limit its ability to form a stable helical structure as multiple aromatic side chains on one helix face have been previously shown to significantly stabilize peptoid helix formation.

	In turning to the KBP 2.20 derived ligands, it was once again observed that the l- and d-peptides bound with essentially identical affinities (KDs of 33 µM and 28 µM, respectively (Figure 10)). The ß-peptide in this series of ligands bound similarly, though with perhaps slightly better affinity (KD 20 µM), when compared to the original KBP 2.20 peptide (Figure 10). However, in comparing the two ligand series, a distinct difference arose in the behavior of the peptoids, with the KBP 2.20 derived peptoid being the highest affinity ligand for the CBP KIX domain examined in this study (KD 5 µM (Figure 10)). The overall greater hydrophobicity of the peptoid ligand may play a role in its higher affinity for the binding surface of the KIX domain, as its backbone possesses a series of aliphatic methylene groups that may allow for an increased hydrophobic surface area occlusion upon binding. The binding results with the KBP 1.66- and KBP 2.20-derived peptides and peptidomimetics represent a direct demonstration that the same functionality can be displayed in a variety of ways while allowing a ligand to retain binding affinity for a target surface. The arginine and proline mutagenesis results with the KBP 2.20 peptide suggest that a radial array of that functionality is an important element in effectively interacting with activation domain binding surfaces. Overall, these results suggest that a number of peptidomimetic scaffolds and organic small molecules may be able to interact with the surfaces of transcriptional coactivators and to function as transcriptional activation domains.
Bibliography Type:	Description: (Last Updated:)
Articles in Peer-reviewed Journals	Rowe SP, Casey RJ, Brennan BB, Buhrlage SJ, Mapp AK. "Transcriptional upregulation in cells mediated by a small molecule." J. Am. Chem. Soc. Epub 2007 Aug 11. PMID: 17691790 , Aug-2007